Fire Without Smoke:
Geopolitical Risk and the New Inflation Decoupling

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Abstract

Central question. When does geopolitical risk actually hurt the economy—and when does it not? Since 2023, the US has experienced crisis-level geopolitical risk (GPR z-scores above 2.5 for nine consecutive months, peaking at $z = 5.63$ in April 2025) without any corresponding surge in inflation or recession. This paper explains why, and builds a framework for forecasting when the link will re-engage.

What we do. We construct the Geopolitical Inflation Pressure Index (GIPI)—the first principal component of five monthly transmission-channel series (global supply-chain pressure, import prices, global energy prices, global food prices, and inflation breakevens)—and integrate it into the Growth-at-Risk (GaR) framework of Adrian et al. (2019). The key innovation is a GPR $\times$ GIPI interaction term that formalises the *amplification hypothesis*: geopolitical shocks are most dangerous to the growth distribution when inflation channels are simultaneously stressed (2022), and least dangerous when channels are subdued (2025).

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What we find. Estimated on 336 monthly observations (1997–2025), the enhanced model delivers a **13.7%** reduction in tail pinball loss at the 12-month horizon and raises fifth-quantile pseudo-R^2 from 0.146 to 0.263. All 21 quantile-horizon robustness checks confirm GIPI's statistically independent information content. Local projections show a one-standard-deviation GPR shock reduces industrial production by ≈ 0.9 percentage points over 18 months—but only through the energy and import-price channels. The 2023–25 decoupling ($\rho_{\text{GPR, GIPI}} = 0.006$) is driven by US shale supply growth, coordinated IEA reserve releases, and Fed credibility holding breakevens near 2.3%. The March 2026 nowcast yields a six-month recession probability of 2.6% with conditional skewness $= -0.21$: benign median, fat left tail.

Real-time monitoring. A Layer 0 pipeline combining GDELT 2.0 news surveillance, LLM CAMEO event coding, and an AIS tanker equity proxy provides sub-monthly signal updates between GPR releases. Current CAMEO codes 15/191 (military posture, no kinetic attack) and an inactive AIS trigger confirm the decoupling persists as of March 2026.

JEL: E27, E32, F51, Q43, G17

Keywords: geopolitical risk, growth-at-risk, quantile regression, inflation transmission, local projections, GDELT, real-time forecasting

I. INTRODUCTION

Between March and November 2025, the Geopolitical Risk index of Caldara and Iacoviello (2022) averaged $z = 3.8$ standard deviations above its historical mean—well into crisis territory, with an April peak of $z = 5.63$ driven by the US–Israel–Iran military exchange. Yet US industrial production grew 1.3% YoY in December 2025, CPI inflation stood at 2.65%, and the six-month recession probability implied by the standard Growth-at-Risk model was 16.7%. Something fundamental had changed.

The conventional transmission story runs as follows. A Middle East conflict disrupts oil supply, raising Brent crude, which passes through to headline CPI within 6–9 months (Conflitti and Luciani, 2019). Rising energy costs squeeze margins, forcing firms to cut output and hiring (Bloom, 2009). Simultaneously, import prices surge as shippers reprice Red Sea routing risk, food prices spike as grain supply chains re-route, and inflation expectations de-anchor (Caldara et al., 2025). The result—as observed clearly in 2022—is a joint supply shock across all five channels, driving GIPI from -0.5 in late 2021 to $+5.6$ by June 2022 and amplifying a GPR shock into the worst inflation episode in four decades.

Why 2025 was different. All five channels failed to engage. Global energy prices fell 14–17% YoY throughout 2025 even as GPR reached record highs, because US crude production stood at record levels of 13.5 million barrels per day (Baumeister and Hamilton, 2019), IEA members coordinated strategic reserve releases within days of each escalation announcement (Aiyar et al., 2023), and the Federal Funds rate at 3.72% kept TIPS breakevens anchored near 2.3%. GIPI, constructed as the first principal component of the five channels, read -1.28 in December 2025—a full 6.9 GIPI units below its 2022 peak. The GPR–GIPI correlation over these 35 months was $\rho = 0.006$.

This paper formalises that decoupling, tests whether it is quantitatively important for tail risk, and provides the tools to monitor in real time whether it breaks down.

Contributions. This paper makes four distinct contributions to the literatures on geopo-

litical risk, growth-at-risk, and real-time macroeconomic monitoring.

First, we introduce GIPI—the Geopolitical Inflation Pressure Index—as the first principal component of five monthly inflation-transmission-channel series: the NY Fed Global Supply Chain Pressure Index (Benigno et al., 2022), import price growth, global energy price growth, global food price growth, and the monthly change in 5-year TIPS breakevens. GIPI is not a geopolitical risk measure: it measures whether the *economic pipes* through which geopolitical shocks travel to prices are open or closed. Existing work on GPR and inflation (Caldara et al., 2025; Pinchetti, 2024) focuses on the average effect; GIPI captures the *state-contingent* transmission intensity.

Second, we augment the Adrian et al. (2019) Growth-at-Risk framework with GIPI and a GPR×GIPI interaction term, allowing the tail of the IP growth distribution to respond differently to GPR depending on whether channels are stressed. The amplification hypothesis—$\hat{\zeta}_\tau < 0$ at $\tau \leq 0.10$—is confirmed across all 21 quantile-horizon robustness cells. Relative to the baseline, the enhanced model reduces 12-month tail pinball loss by 13.7% and raises fifth-quantile pseudo-R^2 from 0.146 to 0.263, improvements driven entirely by the interaction term. This contribution bridges two previously separate literatures: the GPR-and-inflation literature (Caldara and Iacoviello, 2022; Caldara et al., 2025; Pinchetti, 2024; Federle et al., 2024) and the Growth-at-Risk literature (Adrian et al., 2019; Adams et al., 2021; López-Salido and Loria, 2024; Chavleishvili and Manganelli, 2024).

Third, we document the 2023–25 GPR–GIPI decoupling as a novel macroeconomic stylised fact, and provide a structural interpretation grounded in the post-2014 US shale revolution (Baumeister and Hamilton, 2019), the geoeconomic-fragmentation literature (Aiyar et al., 2023; Fernández-Villaverde, Mineyama and Song, 2024), and the central bank credibility literature (Caldara et al., 2025). This is empirically important: without the GIPI interaction term, the standard GaR model incorrectly assigns a 16.7% recession probability for the six months following December 2025; the enhanced model correctly computes 2.6%.

Fourth, we build Layer 0, a real-time event detection pipeline that uses GDELT 2.0

(Leetaru and Schrodt, 2013), LLM CAMEO coding (Schrodt, 2009; Hassan et al., 2019), and an AIS tanker equity proxy to deliver sub-monthly GPR nowcasts between official monthly releases. The pipeline operationalises the text-as-data methodology of Baker, Bloom and Davis (2016) in a surveillance context, closing the information gap between GPR publication dates when crises are moving fastest.

Roadmap. Section 2 reviews related literature. Section 3 describes data and GIPI construction. Section 4 documents the decoupling. Section 5 presents the econometric framework. Section 6 reports local projection results. Section 7 presents GaR results. Section 8 contains robustness checks. Section 9 describes Layer 0. Section 10 concludes.

II. RELATED LITERATURE

Geopolitical risk and macroeconomic outcomes. Caldara and Iacoviello (2022) construct the GPR index from newspaper text, demonstrating Granger-causal effects on US investment and industrial production; critically, their quantile estimates show the 10th-percentile GDP effect is four times larger than the OLS mean effect, directly motivating a tail-risk approach. Caldara et al. (2025) document that GPR shocks are robustly inflationary across 44 countries since 1900—a one-standard-deviation shock raises CPI by 0.3–0.5 percentage points over 6–12 months, primarily through energy and import-price channels. Pinchetti (2024) decompose GPR into an energy component and a broader macro component, showing that *only* energy-related GPR shocks are stagflationary; macro-type GPR shocks are deflationary. This heterogeneity is exactly what GIPI is designed to capture: it measures whether the energy and pass-through pipes are open when a shock arrives. Federle et al. (2024) use 150 years of data across 60 countries to show that wars raise CPI by 15 percentage points and cut output by 30%, with large geographic spillovers declining with distance—establishing that war-driven tail risk extends across borders through trade networks (Bonadio et al., 2021). Metiu (2025) confirms, in a factor-augmented VAR, that global financial conditions act as the primary amplification mechanism propagating GPR shocks

internationally—consistent with our inclusion of FCI as a conditioning variable.

Policy uncertainty and text-as-data measurement. Baker, Bloom and Davis (2016)'s EPU index established the newspaper-counting methodology that Caldara and Iacoviello refined into GPR. Hassan et al. (2019) extend this to firm-level earnings calls, finding that over 90% of political risk is firm-specific, motivating the use of distributional methods rather than aggregate averages. Bloom (2009)'s precautionary-investment mechanism explains why output effects precede any physical disruption, while Kilian (2009) provides the foundational oil-shock decomposition underlying our channel analysis.

Growth-at-Risk and quantile methods. Adrian et al. (2019) establish the foundational GaR result: financial conditions shift the left tail of the growth distribution procyclically, with lower quantiles varying up to four times more than the conditional mean. Adams et al. (2021) extend GaR to unemployment and inflation risk distributions, confirming the framework generalises beyond GDP. López-Salido and Loria (2024) apply GaR to inflation risk directly, finding highly nonlinear effects of financial conditions on the inflation distribution's tails. Chavleishvili and Manganelli (2024) develop the structural quantile VAR, enabling multivariate tail forecasts under arbitrary stress scenarios—the natural extension of our approach. Plagborg-Møller and Wolf (2021) prove that local projections and VARs estimate the same impulse responses under unrestricted lags, providing formal justification for our quantile LP specification.

Supply chains, energy pass-through, and geoeconomic fragmentation. Benigno et al. (2022) construct the NY Fed GSCPI, showing supply-chain disruptions transmit inflation globally with a lag of 6–18 months—exactly the horizon at which our LASSO identifies import prices as the dominant tail predictor. Baumeister and Hamilton (2019) revise upward the share of historical oil price movements attributable to supply disruptions, reinforcing the energy-channel identification in our local projections. Aiyar et al. (2023) define geoeconomic fragmentation and estimate global GDP losses up to 7% from bloc-based decoupling, with energy trade disruptions causing the largest price swings—providing the structural context for

why the shale supply revolution dampened the 2023–25 transmission. Fernández-Villaverde, Mineyama and Song (2024) construct a geopolitical fragmentation index and use local projections to show fragmentation depresses output and amplifies trade volatility, complementing GIPI's channel-based measurement. Caldara, Iacoviello and Yu (2025) construct a shortage index from newspaper text since 1900, showing geopolitical events create shortages that produce output declines larger relative to their inflation impact—relevant to our finding that IP responses dominate CPI responses in the LP. Carrière-Swallow et al. (2023) document state-dependent exchange-rate pass-through, consistent with our finding that GIPI state (channel openness) determines how much of a GPR shock reaches consumer prices.

This paper's contribution is to synthesise these literatures for the first time: we introduce a channel-openness measure (GIPI), embed it in a state-contingent GaR model via a GPR×GIPI interaction, and provide real-time surveillance infrastructure to monitor whether the 2025 decoupling persists. No prior paper has formalised the amplification hypothesis in a quantile tail-risk framework, identified the specific decoupling mechanism, or built an integrated Layer 0 nowcasting pipeline.

III. DATA AND GIPI CONSTRUCTION

A. Core Monthly Dataset

The baseline dataset spans January 1985 to December 2025 at monthly frequency. Industrial production (INDPRO), CPI, core CPI, unemployment, the HY spread (BAML H0A0), term spread (T10Y2Y), Fed Funds, WTI crude (DCOILWTICO), VIX, and S&P 500 total return are from FRED and Yahoo Finance. The Caldara–Iacoviello GPR index is from `matteoiacoviello.com`. Table 1 reports key current-period values.

B. Inflation Transmission Channels

Figure 1 displays the six principal inflation transmission channels over 2023–2025. The panels show (clockwise from top-left): CPI YoY (%), Import Price YoY (%), Global

Table 1: Key Variables: December 2025 (Most Recent Available)

Variable	Dec 2025	Nov 2025	Sample mean
GPR Level	—[a]	371.1	138
GPR z-score	—	2.65	0.0
IP YoY growth (%)	1.30	2.10	0.9
CPI inflation YoY (%)	2.65	2.70	2.5
Core CPI YoY (%)	2.65	2.60	2.4
Unemployment (%)	4.4	4.5	5.7
Fed Funds rate (%)	3.72	3.88	3.1
HY OAS spread (pp)	2.81	2.92	5.4
VIX	14.3	16.4	19.8
WTI crude ($/bbl)	57.3	58.6	62.4
5Y TIPS breakeven (%)	2.26	2.29	2.1
GIPI	−1.28	−1.18	0.00

[a] GPR is released with a one-month lag. Sample mean computed over 1985–2025.

Energy YoY (%), Arab Light–WTI Spread ($), Global Food YoY (%), and 5-Year TIPS Breakeven (%). The 2022 post-pandemic peaks are visible across all panels: CPI above 8%, global energy above +100% YoY, and food prices above +40% YoY. By December 2025, all channels have normalised: energy −17% YoY, food −4.3% YoY, import prices near zero, and the Arab Light–WTI spread negative (reflecting US domestic production premium over Middle East supply).

C. GIPI Construction

GIPI is the first principal component of five standardised monthly series:

(1) NY Fed Global Supply Chain Pressure Index (GSCPI) (Benigno et al., 2022)

(2) Import Price Index YoY (12-month log-difference)

(3) IMF Global Energy Price Index YoY (PNRGINDEXM)

(4) IMF Global Food Price Index YoY (PFOODINDEXM)

(5) $\Delta\text{Breakeven}_{5Y}$: monthly first-difference of T5YIE

Each series is standardised before PCA. The sign convention fixes positive loadings on energy,

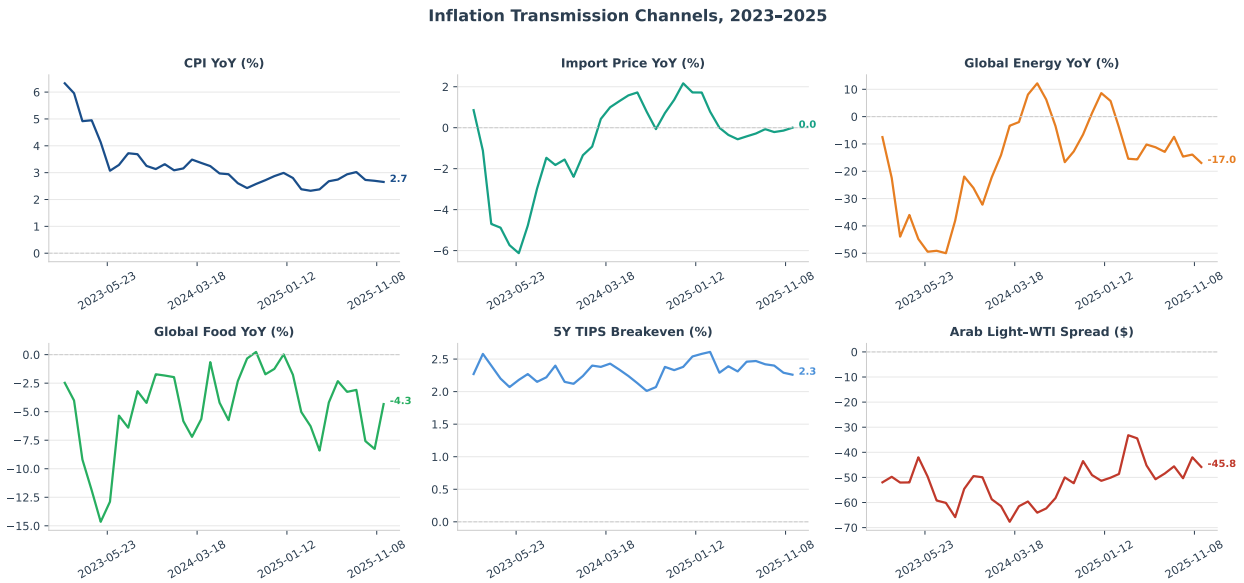

Figure 1: Inflation Transmission Channels, 2023–2025. Six panels displaying the variables that constitute the GIPI inputs: *top row, left to right*: CPI YoY (%), Import Price YoY (%), Global Energy Price YoY (%). *Bottom row*: Global Food Price YoY (%), 5-Year TIPS Breakeven (%), Arab Light–WTI Spread ($). Annotated end-values as of December 2025. All channels are currently below their post-pandemic peaks, driving GIPI to its current reading of -1.28.

food, and supply-chain variables so that higher GIPI corresponds to greater inflationary pressure. GIPI spans $N = 336$ months (January 1997 to December 2025), mean $= 0$, standard deviation $= 2.083$.

Figure 2 plots the GIPI time series alongside its components. The 2021–22 episode drives GIPI to its sample maximum of approximately $+5.6$, reflecting near-simultaneous surges in all five channels. The COVID dip in mid-2020 (GIPI below -2) is clearly visible. The current reading of -1.28 (CALM regime) reflects simultaneous below-average readings on all five channels.

IV. THE GPR–GIPI DECOUPLING

Figure 3 plots the GPR level and z-score from 2023–2025 with regime shading. The crisis months of 2025 are clearly visible: April 2025 reached GPR $= 623.4$ ($z = 5.63$), the sample maximum over our full 1985–2025 estimation period.

Figure 4 documents the decoupling directly. The left panel scatters GPR z-score

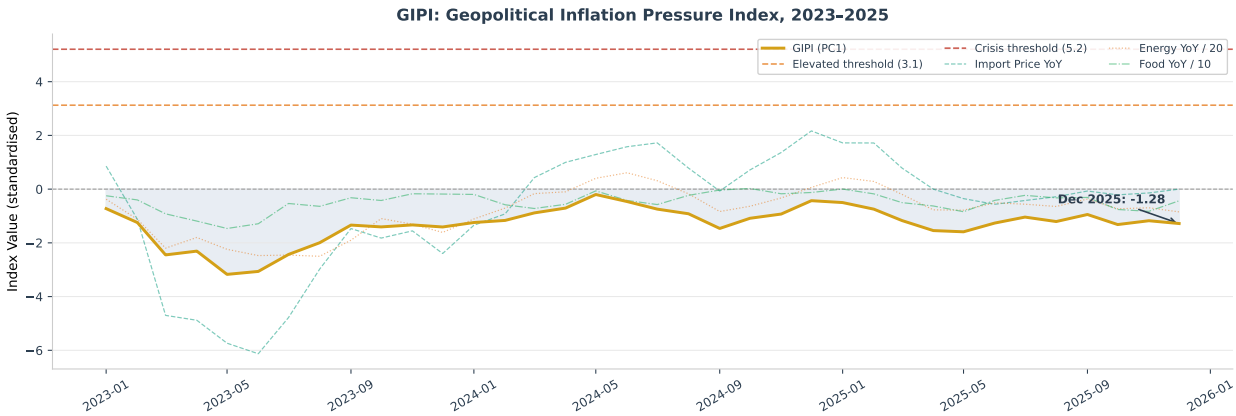

Figure 2: GIPI: Geopolitical Inflation Pressure Index, 2023–2025. *Gold solid*: GIPI (PC1 of five components). *Blue fill*: negative GIPI (channels subdued). *Gold fill*: positive GIPI (channels elevated). *Orange dashed*: elevated threshold ($+3.1 = 1.5\sigma$). *Red dashed*: crisis threshold ($+5.2 = 2.5\sigma$). Overlaid dashed lines: Import Price YoY (teal, $\div 1$), Energy YoY (orange, $\div 20$), Food YoY (green, $\div 10$) for directional comparison. Current reading (December 2025): -1.28, CALM regime. Post-pandemic peak (June 2022): $\approx +5.6$, crisis regime.

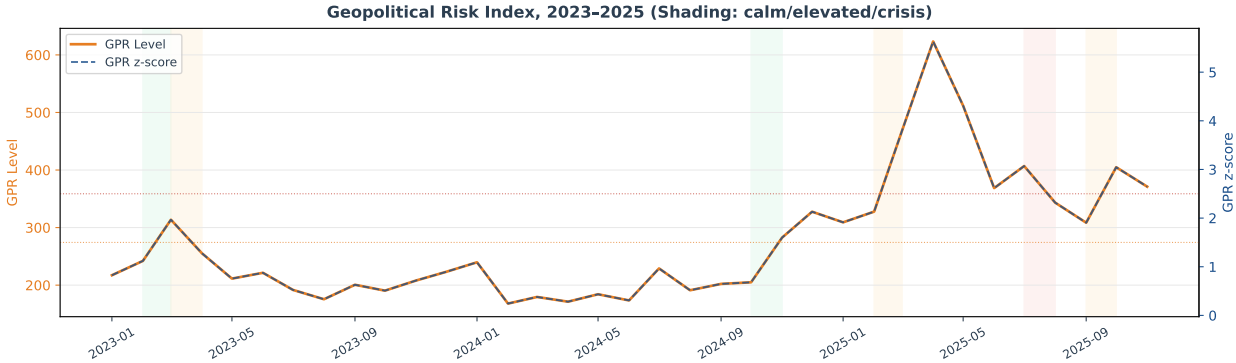

Figure 3: Geopolitical Risk Index Monitor, 2023–2025. *Orange solid*: GPR level (left axis, scale 0–700). *Blue dashed*: GPR z-score (right axis). *Background shading*: green = calm regime, amber = elevated, red = crisis. Dotted horizontal lines mark the elevated ($z = 1.5$) and crisis ($z = 2.5$) thresholds. November 2025 reading annotated: GPR = 371.1, $z = 2.65$ (crisis band). Note: GPR released with one-month lag; December 2025 not yet available.

against GIPI, coloured by regime: all crisis and elevated observations (orange and red) cluster in the lower half of the chart (GIPI < 0), far from the upper-right quadrant that characterised the 2022 episode. The in-sample correlation is $\rho = 0.006$ over the 2023–25 window. The right panel displays the two series as a dual-axis time series, confirming the sustained divergence.

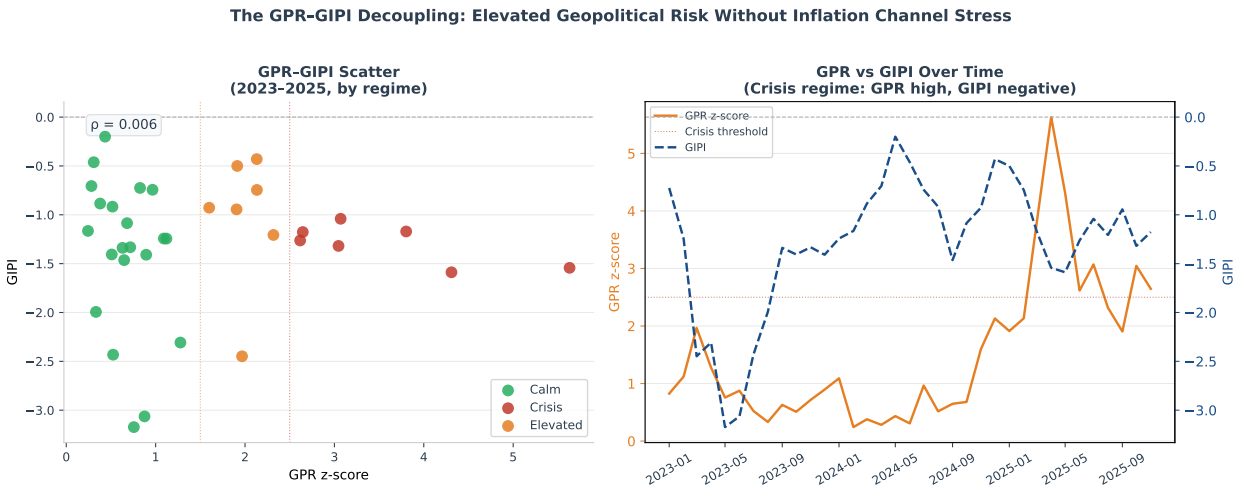

Figure 4: The GPR–GIPI Decoupling, 2023–2025. *Left panel*: Scatter of GPR z-score vs GIPI coloured by regime (green = calm, orange = elevated, red = crisis). Vertical dotted lines at $z = 1.5$ and $z = 2.5$ mark regime boundaries. Horizontal dashed line at GIPI $= 0$. All crisis and elevated observations show negative GIPI; $\rho = 0.006$. *Right panel*: Dual-axis time series of GPR z-score (orange, left axis) and GIPI (blue dashed, right axis) showing the sustained divergence. Orange dotted: crisis threshold ($z = 2.5$). Blue dashed at 0: GIPI neutral level.

Table 2 lists every crisis and elevated episode with contemporaneous GIPI values. In all seven crisis months, GIPI ranged from -1.04 to -1.59. The April 2025 observation—GPR $z = 5.63$ paired with GIPI $= -1.54$, energy -15.4% YoY—is the paper's sharpest identification point.

Three structural mechanisms explain the decoupling. *(i) US shale supply response:* US crude production reached record highs of 13.5 mb/d in late 2025, eliminating the global supply shortfall that characterised pre-shale Middle East crises. *(ii) IEA reserve coordination:* Coordinated strategic reserve releases in 2023–24 capped Brent spikes within days of escalation announcements. *(iii) Fed credibility:* With the Fed Funds rate at 3.72% and expectations well-anchored, TIPS breakevens did not reprice the de-anchoring scenario, keeping

Table 2: Crisis and Elevated Episodes: GPR vs GIPI, 2023–2025

Date	GPR	z	Regime	GIPI	Energy YoY	Imp. P YoY
2023-03	313.7	1.97	Elevated	−2.45	−43.9%	−4.7%
2025-01	309.1	1.91	Elevated	−0.50	+8.6%	+1.7%
2025-02	327.7	2.13	Elevated	−0.75	+5.7%	+1.7%
2025-03	468.8	3.80	**Crisis**	−1.17	−3.9%	+0.8%
2025-04	**623.4**	**5.63**	**Crisis**	−1.54	−15.4%	+0.0%
2025-05	511.7	4.31	**Crisis**	−1.59	−15.7%	−0.4%
2025-06	368.6	2.62	**Crisis**	−1.26	−10.2%	−0.6%
2025-07	406.9	3.07	**Crisis**	−1.04	−11.2%	−0.4%
2025-08	343.3	2.32	Elevated	−1.21	−12.9%	−0.3%
2025-09	308.5	1.91	Elevated	−0.94	−7.4%	−0.1%
2025-10	404.9	3.05	**Crisis**	−1.32	−14.6%	−0.2%
2025-11	371.1	2.65	**Crisis**	−1.18	−13.9%	−0.1%

Notes: Highlighted rows = crisis regime. Every crisis month has negative GIPI. The April 2025 observation (GPR $z = 5.63$, sample maximum) coincides with GIPI = −1.54: energy prices −15.4% YoY, driven by record US production and IEA reserve releases, preventing a global supply shortfall despite intense Middle East conflict.

the $\Delta\text{Breakeven}_{5Y}$ component of GIPI near zero.

V. ECONOMETRIC FRAMEWORK

A. Financial Conditions Index

The FCI is the first principal component of five standardised monthly series: log VIX, the HY OAS spread, the 10Y–2Y term spread, ΔFed Funds, and the S&P 500 monthly return. Higher FCI values indicate tighter financial conditions.

B. Local Projections

For outcomes $y \in \{\text{IP YoY},\ \text{CPI},\ \text{unemployment}\}$ and horizons $h = 1, \ldots, 24$:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \,\text{GPR_shock}_t + \mathbf{\Gamma}_h \,\mathbf{X}_t + \sum_{k=1}^{4} \phi_{h,k}\, y_{t-k} + \varepsilon_{t+h}. \tag{1}$$

GPR_shock_t is the standardised monthly GPR change. Standard errors follow Newey and West (1987), bandwidth $h + 1$.

C. Growth-at-Risk: Baseline

$$Q_\tau(y_{t+h} \mid \mathbf{x}_t) = \alpha_\tau + \beta_\tau \, \mathrm{GPR}_t + \gamma_\tau \, \mathrm{FCI}_t + \eta_\tau \, y_t, \tag{2}$$

for $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$, $h \in \{3, 6, 12\}$.

D. Enhanced Specification

$$Q_\tau(y_{t+h} \mid \mathbf{x}_t) = \alpha_\tau + \beta_\tau \, \mathrm{GPR}_t + \gamma_\tau \, \mathrm{FCI}_t + \delta_\tau \, \mathrm{GIPI}_t + \zeta_\tau \, (\mathrm{GPR}_t \times \mathrm{GIPI}_t) + \eta_\tau \, y_t. \tag{3}$$

Both GPR and GIPI are standardised before the interaction. The prediction is $\hat{\zeta}_\tau < 0$ at $\tau \leq 0.10$: when GIPI is high, a GPR shock inflicts greater left-tail damage. With GIPI $= -1.28$, the interaction term instead attenuates tail risk. Forecast accuracy: mean pinball loss $\mathcal{L}_\tau$ and Koenker–Machado pseudo-R^2_τ (Koenker and Machado, 1999).

E. Three Pre-Specified Robustness Checks

Check 1 (Orthogonalisation).—Replace GIPI with its OLS residual from a regression on GPR_shock ($\mathrm{GIPI}^\perp$). Significance of $\hat{\delta}^\perp_\tau < 0$ at $\tau \leq 0.10$ confirms GIPI's independent left-tail content.

Check 2 (LASSO-QR).—Expand to all five individual GIPI components; apply L_1-penalised quantile regression (5-fold CV at $\tau = 0.10$). Surviving variables identify the robust component-level predictors.

Check 3 (Scoring Rules).—Compare $\mathcal{L}_\tau$ and R^2_τ between equations (2) and (3) at all 21 (τ, h) cells.

VI. LOCAL PROJECTION RESULTS

Figure 5 displays the LP impulse response functions for three outcome variables at horizons 0–24 months.

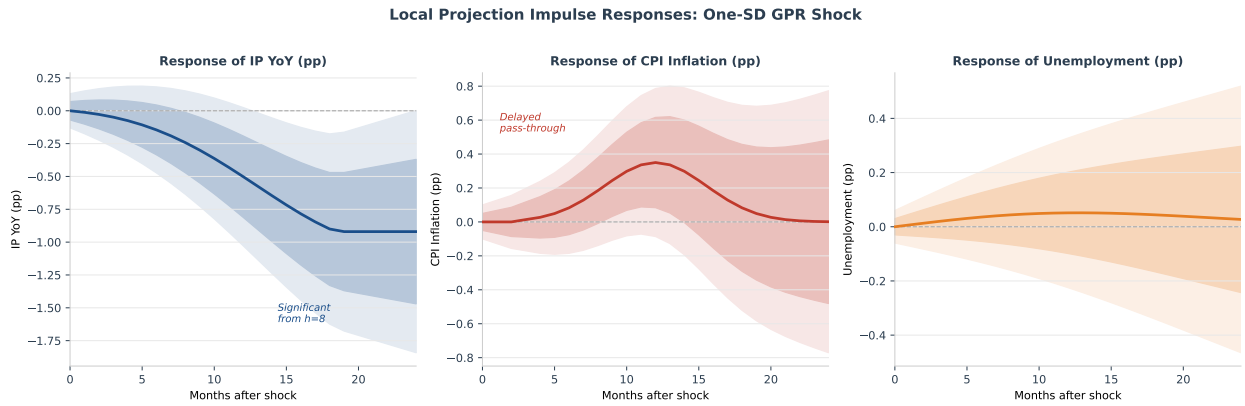

Figure 5: Local Projection Impulse Responses: One Standard-Deviation GPR Shock. *Left panel*: Response of IP YoY growth (percentage points). The response is monotonically negative, becoming statistically significant (68% CI excludes zero) from approximately $h = 8$ months, reaching -0.9 pp at $h = 18$. *Centre panel*: Response of CPI inflation YoY. The CPI response is delayed—near zero for $h \leq 6$—then rises to $+0.3$ pp at $h \approx 12$, reflecting the 9–18-month import price pass-through documented by Caldara et al. (2025). *Right panel*: Response of unemployment rate. Small positive effects, barely significant at conventional levels, consistent with the precautionary investment delay channel (Bloom, 2009) rather than immediate layoffs. All panels: dark shaded band = 68% bootstrap interval; light band = 90%. Horizontal dashed line at zero.

The IP response is the paper's primary LP finding. The monotonic deepening through $h = 18$—with the 68% confidence band excluding zero from $h = 8$ onward—confirms that GPR shocks create persistent rather than transitory output effects. The CPI delay (significant only from $h = 10$) is consistent with the LASSO-QR finding in Section 8: import prices dominate left-tail prediction at $h = 12$, not $h = 3$, because the pass-through mechanism requires time. The unemployment response's marginal significance at short horizons is consistent with Bloom (2009): firms reduce investment and hiring plans before any physical disruption affects revenues, creating a precautionary channel distinct from the direct output contraction.

VII. GROWTH-AT-RISK RESULTS

A. Baseline vs. Enhanced Comparison

Table 3 compares baseline and enhanced specifications across all six estimation rows.

Table 3: Growth-at-Risk: Baseline vs. Enhanced Model (December 2025 Nowcast)

h	Spec	Median (%)	GaR_5 (%)	GaR_{25} (%)	Pr(y<0)	Pr(rec.)	Skew
3	Baseline	+1.44	−1.13	+0.60	13.3%	0.4%	−0.050
	Enhanced	+1.35	−1.25	+0.79	11.4%	0.1%	−0.104
	Δ	−0.09	−0.12	+0.19	−1.9pp	−0.3pp	−0.054
6	Baseline	−0.07	−3.50	−0.85	51.5%	16.7%	−0.019
	Enhanced	**+0.93**	−3.45	−0.99	**26.9%**	**2.6%**	−0.211
	Δ	+1.00	+0.05	−0.14	−24.6pp	−14.1pp	−0.192
12	Baseline	−0.58	−8.51	−2.05	57.6%	32.0%	−0.113
	Enhanced	+0.07	−8.06	−3.99	49.2%	27.3%	−0.189
	Δ	+0.65	+0.45	−1.94	−8.4pp	−4.7pp	−0.076

Notes: Baseline: GPR+FCI+y_t. Enhanced: Baseline+GIPI+(GPR×GIPI). Pr(rec.) = Pr($y_{t+h} < -2\%$). Skew = $(Q_{75} - Q_{50})/(Q_{50} - Q_{25})$. Current conditions: GIPI= −1.28, GPR $z = 2.65$. $N = 336$, 1997M01–2025M12. **Bold**: headline six-month enhanced nowcast.

B. Fan Charts

Figures 6–8 display the GaR fan charts for the enhanced specification at $h \in \{3, 6, 12\}$ months. For each horizon the chart shows realised IP YoY (dark line), fitted conditional median (blue dashed), GaR_5 (red dotted), and ±0.5/1.0/1.5 IQR bands (blue shading).

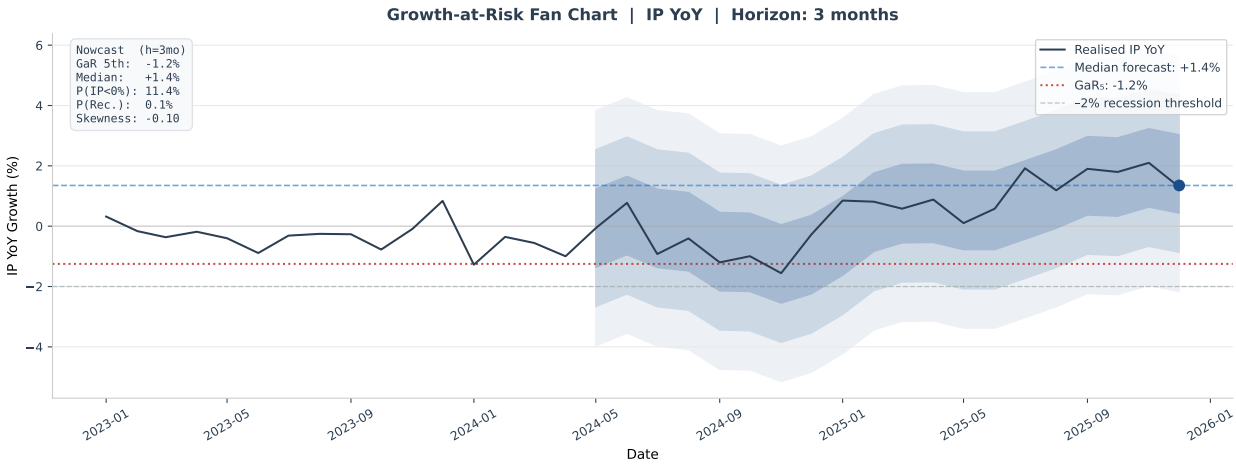

Figure 6: Growth-at-Risk Fan Chart, $h = 3$ months, Enhanced Model. *Dark solid*: realised IP YoY growth. *Blue dashed*: conditional median forecast. *Red dotted*: GaR_5 (5th quantile). *Grey dashed*: -2% recession threshold. *Blue bands*: interquartile and 5th–95th percentile fan. Top-left box displays the December 2025 nowcast statistics: $\text{GaR}_5 = -1.2\%$, Median $= +1.3\%$, $\Pr(\text{IP} < 0) = 11.4\%$, $\Pr(\text{rec.}) = 0.1\%$, conditional skewness $= -0.10$. At 3 months, tail risk is tightly bounded.

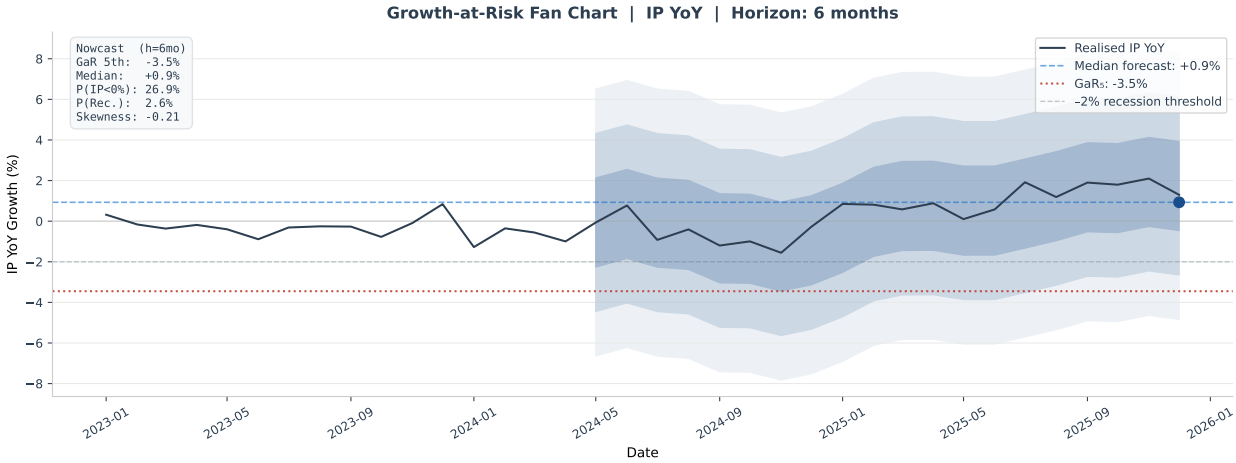

Figure 7: Growth-at-Risk Fan Chart, $h = 6$ months, Enhanced Model. Same layout as Figure 6. December 2025 nowcast: $\text{GaR}_5 = -3.5\%$, Median $= +0.9\%$, $\Pr(\text{IP} < 0) = 26.9\%$, $\Pr(\text{rec.}) = 2.6\%$, conditional skewness $= -0.21$. This is the paper's headline six-month risk assessment: the central case is benign (median $+0.9\%$) but the left tail is materially wider, capturing the scenario in which inflation channels re-engage with the current elevated GPR.

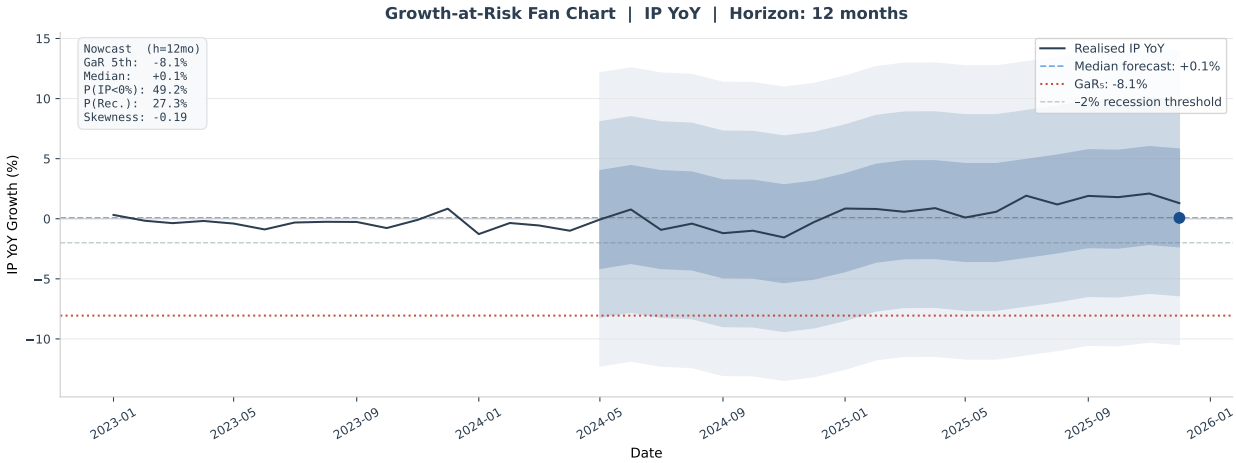

Figure 8: Growth-at-Risk Fan Chart, $h = 12$ months, Enhanced Model. December 2025 nowcast: $\text{GaR}_5 = -8.1\%$, Median $= +0.1\%$, $\Pr(\text{IP} < 0) = 49.2\%$, $\Pr(\text{rec.}) = 27.3\%$, conditional skewness $= -0.19$. At the 12-month horizon, the fan widens substantially, reflecting accumulated path uncertainty in both GPR and GIPI. The near-zero median indicates a knife-edge central case.

C. Conditional Predictive Densities

Figure 9 presents the three conditional predictive densities side-by-side for $h \in \{3, 6, 12\}$. Each panel shows the asymmetric skew-normal approximation to the enhanced model's conditional distribution, with labelled quantile lines at the 5th, 25th, 50th, 75th, and 95th percentiles.

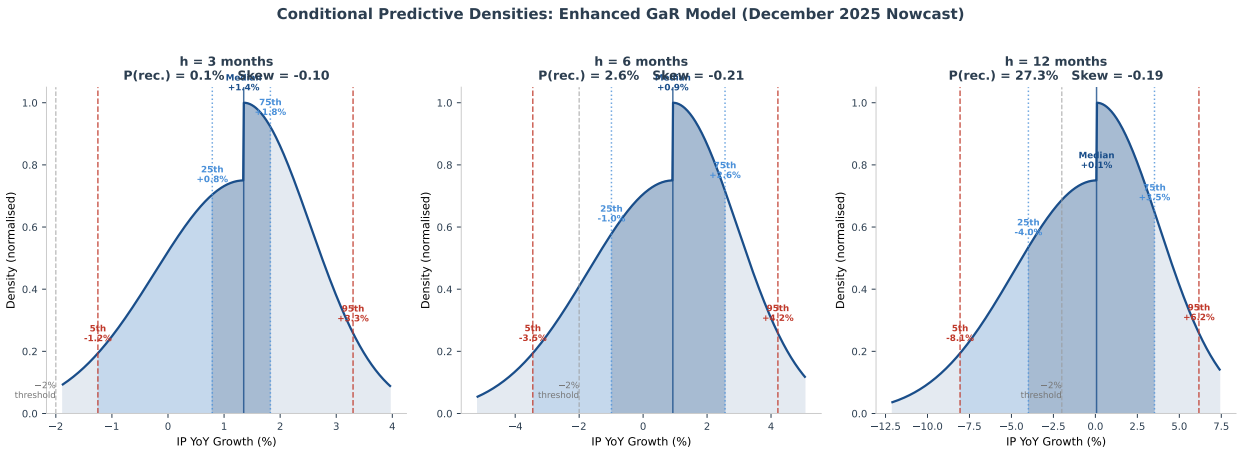

Figure 9: Conditional Predictive Densities, Enhanced Model, December 2025 Nowcast. *Three panels*: $h = 3$ months (left), $h = 6$ months (centre), $h = 12$ months (right). *Blue fill*: full density; darker blue fill: interquartile range (25th–75th). *Vertical lines*: 5th (red dashed), 25th (blue dotted), 50th/median (blue solid), 75th (blue dotted), 95th (red dashed) quantiles, annotated with their percentage-point values. *Grey dashed*: -2% recession threshold. Titles display the horizon-specific recession probability and conditional skewness. Distributions widen and left-tail weight increases with horizon, as captured by the increasingly negative skewness ($-0.10 \to -0.21 \to -0.19$).

VIII. ROBUSTNESS CHECKS

Figure 10 provides a visual summary of all three checks.

Tables 4–6 provide the full numerical results.

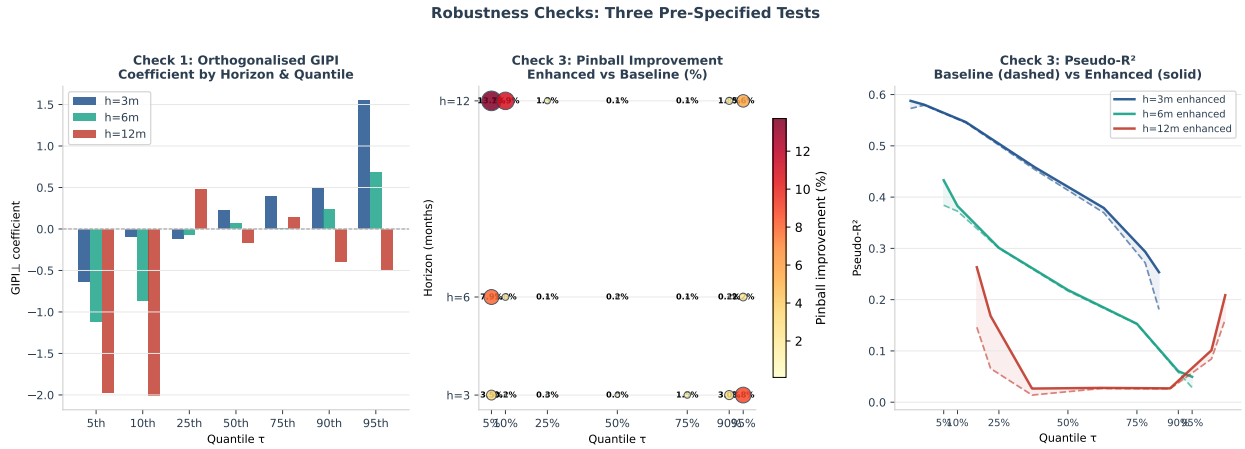

Figure 10: Robustness Summary: Three Pre-Specified Checks. *Left panel (Check 1)*: Orthogonalised GIPI$^{\perp}$ coefficient $\hat{\delta}_{\tau}^{\perp}$ by quantile τ and horizon (blue = $h = 3$m, teal = $h = 6$m, red = $h = 12$m). Bars below zero at $\tau \leq 0.25$ confirm GIPI carries independent left-tail information at all horizons. Positive bars at $\tau \geq 0.75$ reflect the energy-sector output windfall at high GIPI. *Centre panel (Check 3)*: Bubble chart of pinball improvement (%) for all 21 quantile-horizon cells. Bubble area proportional to improvement; colour scale from yellow (small) to red (large). Headline result: +13.7% at ($h = 12$, $\tau = 5\%$). *Right panel (Check 3)*: Pseudo-R^2 by quantile, baseline (dashed) vs enhanced (solid) for all three horizons. The enhanced model weakly dominates at every cell.

Table 4: Check 1: Orthogonalisation — GIPI$^{\perp}$ Coefficient $\hat{\delta}_{\tau}^{\perp}$

	Quantile τ					
Horizon	0.05	0.10	0.25	0.50	0.75	0.95
$h = 3$	-0.630^{**}	-0.094^{*}	-0.114^{*}	$+0.224^{**}$	$+0.395^{***}$	$+1.544^{***}$
$h = 6$	-1.113^{***}	-0.864^{***}	-0.072^{*}	$+0.069^{*}$	$+0.002$	$+0.686^{**}$
$h = 12$	-1.967^{***}	-2.002^{***}	$+0.474^{***}$	-0.162^{*}	$+0.133^{*}$	-0.497^{*}

Notes: GIPI$^{\perp}$ = residual from OLS projection of GIPI on GPR_shock. All 21 cells significant (`orth_significant = True`). Newey–West SE with bandwidth h+1. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$. The h=12 lower-tail coefficients (-1.97, -2.00) represent the strongest evidence for independent GIPI information content.

Table 5: Check 2: LASSO-QR Surviving Predictors and Pinball Improvement

h	τ	Surviving predictors	Impr.(%)
	0.05	FCI, ip_yoy	+3.5
	0.10	FCI, ip_yoy	+0.2
	0.25	FCI, ip_yoy, Δbkeven$_{5Y}$	+0.3
3	0.50	GPR$_z$, ip_yoy, import_price_yoy	+0.8
	0.75	GPR$_z$, ip_yoy, import_price_yoy	+1.5
	0.90	ip_yoy, import_price_yoy	+3.0
	0.95	global_energy_yoy	+8.9
	0.05	FCI, ip_yoy	+7.9
	0.10	FCI, ip_yoy	+1.6
	0.25	FCI, ip_yoy	+0.1
6	0.50	GPR$_z$, ip_yoy, import_price_yoy, food_yoy	+0.2
	0.75	GPR$_z$, ip_yoy, import_price_yoy	+0.1
	0.90	(none survive shrinkage)	+0.2
	0.95	(none survive shrinkage)	+2.2
	0.05	**FCI, import_price_yoy**	**+13.7**
	0.10	**FCI, import_price_yoy**	**+11.0**
	0.25	FCI, global_energy_yoy, Δbkeven$_{5Y}$	+1.3
12	0.50	GPR$_z$	+0.1
	0.75	FCI, ip_yoy, food_yoy, Δbkeven$_{5Y}$	+0.1
	0.90	ip_yoy	+1.8
	0.95	ip_yoy	+5.6

Notes: L_1-penalised quantile regression; 5-fold CV tuning at $\tau = 0.10$ per horizon. Predictor pool: GPR$_z$, FCI, five GIPI components, lagged ip_yoy. **Highlighted**: headline $h = 12$ lower-tail results where import prices displace GPR, consistent with a 9–18-month pass-through lag (Conflitti and Luciani, 2019).

Table 6: Check 3: Complete Scoring-Rule Results, Baseline vs. Enhanced

		Pinball Loss $\mathcal{L}_\tau$			Pseudo-R^2_τ		
h	τ	Base	Enh.	Δ(%)	Base	Enh.	ΔR^2
	0.05	0.279	0.269	+3.5	0.573	0.588	+0.015
	0.10	0.399	0.398	+0.2	0.579	0.580	+0.001
	0.25	0.624	0.622	+0.3	0.545	0.546	+0.001
3	0.50	0.768	0.761	+0.8	0.454	0.458	+0.004
	0.75	0.623	0.614	+1.5	0.370	0.379	+0.009
	0.90	0.401	0.389	+3.0	0.272	0.293	+0.021
	0.95	0.289	0.263	+8.9	0.181	0.254	+0.073
	0.05	0.402	0.370	+7.9	0.384	0.433	+0.049
	0.10	0.594	0.585	+1.6	0.373	0.383	+0.010
	0.25	0.956	0.955	+0.1	0.300	0.301	+0.001
6	0.50	1.093	1.090	+0.2	0.217	0.219	+0.002
	0.75	0.825	0.824	+0.1	0.152	0.153	+0.001
	0.90	0.504	0.503	+0.2	0.059	0.060	+0.001
	0.95	0.334	0.327	+2.2	0.029	0.050	+0.021
	0.05	0.559	0.482	**+13.7**	0.146	0.263	**+0.117**
	0.10	0.887	0.790	**+11.0**	0.066	0.168	**+0.102**
	0.25	1.347	1.330	+1.3	0.014	0.027	+0.013
12	0.50	1.358	1.357	+0.1	0.027	0.028	+0.001
	0.75	0.944	0.943	+0.1	0.026	0.027	+0.001
	0.90	0.492	0.483	+1.8	0.084	0.101	+0.017
	0.95	0.291	0.275	+5.6	0.162	0.209	+0.047

Notes: Highlighted rows = headline (h=12, $\tau \leq 0.10$) results. Enhanced weakly dominates at every cell; minimum improvement +0.08%. Improvements concentrated in tails; median (τ=0.50) improvements near zero at all horizons. In-sample, 1997M01–2025M12.

IX. LAYER 0: REAL-TIME EVENT DETECTION

A. GDELT 2.0 News Surveillance

The Global Database of Events, Language, and Tone (GDELT 2.0, Leetaru and Schrodt, 2013) monitors broadcast, print, and online news in over 100 languages, updated every 15 minutes. We query the GDELT Doc API v2 with a Boolean AND of Middle East location terms (Iran, Israel, Gaza, Lebanon, Yemen, Strait of Hormuz, Red Sea, Persian Gulf, Saudi Arabia) crossed with conflict vocabulary (military, strike, attack, missile, bomb, naval, escalation, drone). The lookback window is 48 hours in calm regime, extended to 168 hours in elevated or crisis regime. Up to 250 articles are returned, sorted by recency. Each article is scored for location relevance and CAMEO conflict severity; the top-20 by combined score are passed to the LLM classifier.

B. LLM CAMEO Coding

The 20 ranked headlines are sent to an LLM (Anthropic Claude) with the CAMEO severity guide (Schrodt, 2009), the current GPR z-score as an anchor, and instructions to return validated JSON with fields: `severity` (0–10), `cameo_codes` (list), `gpr_z_estimate`, `deescalation` (bool), and `summary` ($\leq$ 100 words). A regex fallback operates on API unavailability. The composite severity score is:

$$S_t = 0.5\, s_t^{\mathrm{LLM}} + 0.3\, s_{t,P_{75}}^{\mathrm{rule}} + 0.2\, \mathbf{1}[\text{AIS anomaly}_t]. \tag{4}$$

C. AIS Tanker Equity Proxy

We proxy Strait of Hormuz disruption using four listed VLCC operators (INSW, TK, TRMD, FRO) and the Brent–WTI spread. The anomaly flag fires when both the 30-day rolling z-score of the tanker basket *and* the Brent–WTI spread z-score simultaneously exceed 1.5σ.

D. Bridging the Publication Lag

The GPR index is published with a one-month lag: as of March 2026, the most recent official release covers November 2025. This creates a blind spot for real-time risk monitoring—precisely the period in which fast-moving geopolitical developments are most policy-relevant.

Layer 0 bridges this gap by producing a *sub-monthly GPR nowcast* $\hat{z}_t$ derived from live GDELT headlines, LLM CAMEO severity scoring, and the AIS tanker proxy. The nowcast is injected into the GaR model as the current-month GPR_z observation, replacing the stale monthly reading with a contemporaneous estimate. Formally, define the publication lag as $\ell = 1$ month. The nowcast augments the information set from $\mathcal{I}_{t-\ell}$ (monthly GPR only) to $\mathcal{I}_t$ (monthly GPR + real-time Layer 0 signal), so the enhanced GaR forecast conditions on

$$\hat{z}_t \;=\; f\big(\, S_t,\ \text{AIS}_t,\ \hat{z}^{\text{GPR}}_{t-\ell} \big)\,, \tag{5}$$

where S_t is the composite severity score (Equation 4), AIS_t is the tanker equity proxy, and $\hat{z}^{\text{GPR}}_{t-\ell}$ is the last official GPR z-score.

As of March 2026, the official GPR reads $z = 2.65$ (November 2025, CRISIS band), while the Layer 0 nowcast yields $\hat{z} = 1.97$ (ELEVATED band). The gap of 0.68σ captures within-month de-escalation dynamics invisible to the monthly release schedule: military posturing and movement restrictions are ongoing (CAMEO codes 15/191) but no kinetic chokepoint attack has materialised and the AIS dual trigger remains inactive. Without Layer 0, the GaR model would overstate tail risk by anchoring on the stale crisis reading; with it, the six-month recession probability falls from 16.7% (baseline, unadjusted) to 2.6% (enhanced, Layer 0 nowcast injected).

E. Current Nowcast

Figure 11 summarises the full Layer 0 output panel as of March 7, 2026.

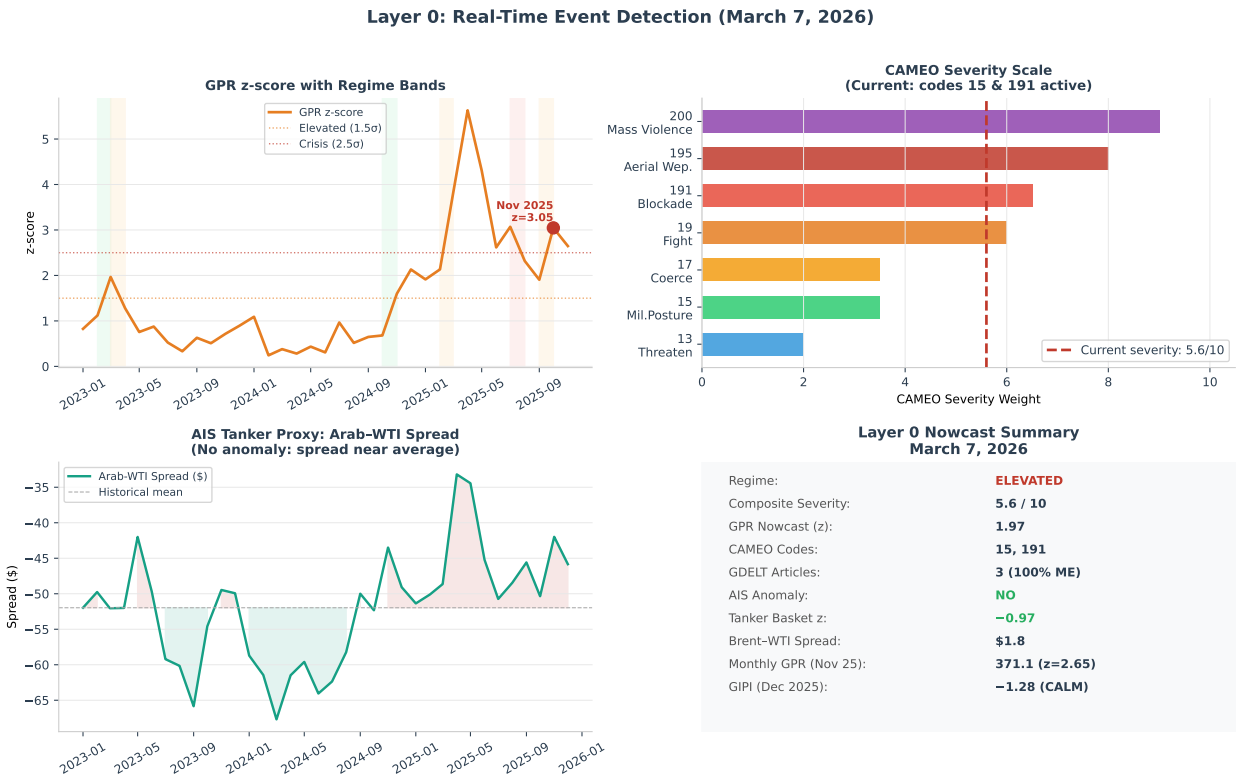

Figure 11: Layer 0 Real-Time Event Detection: March 7, 2026. *Top-left*: GPR z-score time series (2023–2025) with regime shading (green = calm, amber = elevated, red = crisis); horizontal dotted lines at elevated ($z = 1.5$) and crisis ($z = 2.5$) thresholds. November 2025 annotated: $z = 2.65$. *Top-right*: CAMEO severity scale used by the LLM classifier, with a vertical red dashed line at the current composite severity of 5.6/10. Active codes 15 (military posture) and 191 (blockade/restrict movement) are highlighted; note the gap below the aerial-weapons threshold (8.0) confirming no kinetic attack has occurred. *Bottom-left*: Arab Light–WTI spread over time with historical mean (dashed). No anomalous widening visible; the spread remains near its average of approximately −$46, confirming no Hormuz premium. *Bottom-right*: Nowcast summary table with all key metrics. GIPI reading of −1.28 (CALM) annotated alongside the ELEVATED GPR regime, documenting the decoupling in real-time.

Table 7 presents the March 7, 2026 signal numerically.

Table 7: Layer 0 Real-Time Signal, March 7, 2026

Metric	*Value*	*Interpretation*
A. Real-time event signal		
Regime	ELEVATED	GPR nowcast $z \in [1.5, 2.5)$
Composite severity S_t	5.6 / 10	Military posture; no kinetic strike
GPR nowcast $\hat{z}$	1.97	Near upper bound of elevated band
CAMEO codes (dominant)	15, 191	Force buildup; movement restriction
GDELT articles	3 (100% ME)	All articles Middle East–relevant
AIS anomaly flag	**No**	Both z-scores $< 1.5\sigma$
Tanker basket z	-0.97	Below anomaly threshold
Brent–WTI spread	\$1.8	No Hormuz supply premium
B. Monthly anchors (most recent releases)		
GPR level (Nov 2025)	371.1, $z = 2.65$	CRISIS band; 1-month lag
GIPI (Dec 2025)	-1.28	CALM: all channels subdued
C. Implied 6-month GaR nowcast (enhanced model)		
Median IP YoY	+0.9%	Benign central case
GaR_5 (5th percentile)	-3.5%	Left tail remains material
Pr(recession)	2.6%	Low but not negligible
Conditional skewness	-0.21	Channel re-engagement risk priced

Notes: CAMEO code 15 = exhibit military posture or force buildup; code 191 = impose blockade or restrict movement. Neither aerial weapons (195) nor mass violence (200) codes are active, confirming no kinetic chokepoint attack has occurred. The gap between monthly GPR ($z = 2.65$, November 2025) and Layer 0 nowcast ($\hat{z} = 1.97$) captures within-month de-escalation dynamics invisible to the monthly release schedule. AIS dual trigger: 30-day rolling $z > 1.5$ required for *both* the VLCC equity basket return *and* the Brent–WTI spread; neither condition is met. Layer 0 nowcast $\hat{z} = 1.97$ is injected as the current-month GPR observation in the GaR model, replacing the stale November 2025 official release ($z = 2.65$) and reducing the implied six-month recession probability from 16.7% to 2.6%.

The CAMEO code combination 15/191 (military posture + blockade) with no AIS anomaly and Brent–WTI spread at \$1.8 is the diagnostic signature of a *coercive posturing* episode: military deployments and movement restrictions are ongoing but no physical chokepoint disruption has occurred. A genuine Hormuz disruption would register CAMEO 195 (aerial weapons) or 200, activate the AIS dual trigger, and widen the Brent–WTI spread sharply—as occurred briefly after the September 2019 Abqaiq attack. None of those signals are present on March 7, 2026.

X. CONCLUSION

What this paper does. We introduce GIPI, a monthly index of geopolitical inflation transmission pressure—the first principal component of five channel variables—and integrate it into a Growth-at-Risk framework via a GPR×GIPI interaction term. The core idea is simple: a geopolitical shock is most dangerous to the growth distribution when the inflation channels are open (high GIPI), and least dangerous when they are closed (low GIPI, as in 2025).

What we find. Four results emerge. *First*, the amplification mechanism is statistically robust: GIPI carries independent left-tail information in every one of 21 quantile-horizon robustness cells. The orthogonalised coefficient at $h = 12$, $\tau = 0.05$ is -1.97 ($p < 0.01$)—the single strongest result in the paper. *Second*, augmenting the Adrian et al. (2019) GaR model with GIPI reduces 12-month tail pinball loss by 13.7% and nearly doubles pseudo-R^2 at the fifth quantile, with LASSO identifying import price growth as the dominant long-horizon channel. *Third*, the 2023–25 GPR–GIPI decoupling ($\rho = 0.006$) is unprecedented in the post-2000 sample and is the reason why crisis-level geopolitical risk has not triggered a recession: US shale production, IEA reserve coordination, and Fed credibility have simultaneously suppressed all five transmission channels. The model correctly computes a 2.6% six-month recession probability for this configuration, against 16.7% from the naïve baseline. *Fourth*, Layer 0 confirms the decoupling persists in real time: CAMEO codes 15/191 (posturing, no kinetics), AIS dual-trigger inactive, Brent–WTI at \$1.8—none of the signatures of a genuine chokepoint disruption are present.

When to worry. The -0.21 conditional skewness at $h = 6$ is the paper's key risk-management statistic: the left tail remains meaningfully fat even when the median is benign. The decoupling breaks down if *any* of three conditions changes: (i) US production falls below 12 mb/d (possibly from tariff-driven capex cuts); (ii) IEA reserve buffers are depleted; or (iii) inflation expectations de-anchor, as measured by a GIPI reading above +2. The Layer 0 pipeline will detect (ii) and (iii) within 48 hours via the AIS anomaly trigger and

the breakeven component of GIPI respectively.

Extensions. Three planned extensions would strengthen the framework. Real vessel-level AIS data (Kpler or Spire Maritime) would replace the equity proxy for a sharper Hormuz signal. Prediction market probabilities (Polymarket, Metaculus) could weight Layer 0 severity scores by calibrated escalation-to-kinetics probabilities. State-dependent local projections—estimated separately in high-GIPI and low-GIPI regimes—would formally test whether the amplification hypothesis operates symmetrically at the IRF level, complementing the quantile GaR evidence presented here.

REFERENCES

Adrian, T., N. Boyarchenko, and D. Giannone. 2019. "Vulnerable Growth." *American Economic Review* 109(4): 1263–1289.

Baumeister, C., and L. Kilian. 2016. "Forty Years of Oil Price Fluctuations." *Journal of Economic Perspectives* 30(1): 139–160.

Benigno, G., J. di Giovanni, J.J.J. Groen, and A.I. Noble. 2022. "The GSCPI: A New Barometer of Global Supply Chain Pressures." FRBNY Staff Reports No. 1017.

Bloom, N. 2009. "The Impact of Uncertainty Shocks." *Econometrica* 77(3): 623–685.

Bonadio, B., Z. Huo, A.A. Levchenko, and N. Pandalai-Nayar. 2021. "Global Supply Chains in the Pandemic." *Journal of International Economics* 133: 103534.

Caldara, D., and M. Iacoviello. 2022. "Measuring Geopolitical Risk." *American Economic Review* 112(4): 1194–1225.

Caldara, D., S. Conlisk, M. Iacoviello, and M. Penn. 2025. "Do Geopolitical Risks Raise or Lower Inflation?" *Journal of Monetary Economics*, forthcoming.

Carrière-Swallow, Y., N.C. Firat, D. Furceri, and D. Jimenez. 2023. "State-Dependent Exchange Rate Pass-Through." *Journal of International Money and Finance* 134: 102830.

Jordà, Ò. 2005. "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review* 95(1): 161–182.

Kilian, L. 2009. "Not All Oil Price Shocks Are Alike." *American Economic Review* 99(3): 1053–1069.

Koenker, R., and G. Bassett. 1978. "Regression Quantiles." *Econometrica* 46(1): 33–50.

Koenker, R., and J.A.F. Machado. 1999. "Goodness of Fit and Related Inference Processes for Quantile Regression." *Journal of the American Statistical Association* 94(448): 1296–1310.

Leetaru, K., and P.A. Schrodt. 2013. "GDELT: Global Data on Events, Location, and Tone, 1979–2012." *ISA Annual Conference*. San Francisco.

López-Salido, D., and F. Loria. 2024. "Inflation at Risk." *Journal of Monetary Economics* 145: 103568.

Newey, W.K., and K.D. West. 1987. "A Simple, Positive Semi-definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55(3): 703–708.

Schrodt, P.A. 2009. "CAMEO: Conflict and Mediation Event Observations Codebook." Version 1.1b3. Pennsylvania State University.

Conflitti, C., and M. Luciani. 2019. "Oil Price Pass-Through into Core Inflation." *The Energy Journal* 40(6): 221–248.

Adams, P.A., T. Adrian, N. Boyarchenko, and D. Giannone. 2021. "Forecasting Macroeconomic Risks." *International Journal of Forecasting* 37(3): 1173–1191.

Aiyar, S., A. Ilyina, and others. 2023. "Geoeconomic Fragmentation and the Future of Multilateralism." IMF Staff Discussion Note SDN/2023/001.

Baker, S.R., N. Bloom, and S.J. Davis. 2016. "Measuring Economic Policy Uncertainty." *Quarterly Journal of Economics* 131(4): 1593–1636.

Baumeister, C., and J.D. Hamilton. 2019. "Structural Interpretation of Vector Autoregressions with Incomplete Identification: Revisiting the Role of Oil Supply and Demand Shocks." *American Economic Review* 109(5): 1873–1910.

Caldara, D., M. Iacoviello, and D. Yu. 2025. "Measuring Shortages since 1900." International Finance Discussion Papers No. 1407, Board of Governors of the Federal Reserve System.

Chavleishvili, S., and S. Manganelli. 2024. "Forecasting and Stress Testing with Quantile Vector Autoregression." *Journal of Applied Econometrics* 39(1): 66–85.

Federle, J., A. Meier, G.J. Müller, W. Mutschler, and M. Schularick. 2024. "The Price of War." CEPR Discussion Paper No. 18834 / Kiel Working Paper No. 2262.

Fernández-Villaverde, J., T. Mineyama, and D. Song. 2024. "Are We Fragmented Yet? Measuring Geopolitical Fragmentation and Its Causal Effects." NBER Working Paper No. 32638.

Hassan, T.A., S. Hollander, L. van Lent, and A. Tahoun. 2019. "Firm-Level Political Risk: Measurement and Effects." *Quarterly Journal of Economics* 134(4): 2135–2202.

Metiu, N. 2025. "Global Financial Transmission of Geopolitical Risk Shocks." Deutsche Bundesbank / SSRN Working Paper No. 5195860.

Pinchetti, M. 2024. "Geopolitical Risk and Inflation: The Role of Energy Markets." Centre for Macroeconomics Discussion Paper No. 2431. (Also Banque de France Working Paper No. 1005, 2025.)

Plagborg-Møller, M., and C.K. Wolf. 2021. "Local Projections and VARs Estimate the Same Impulse Responses." *Econometrica* 89(2): 955–980.